

Fourie van Rooyen

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EDUCATION

The University of Alabama | Tuscaloosa, AL

Bachelor of Science in Aerospace Engineering | Expected May 2026 | Major GPA: 3.0

Relevant Coursework: Solid Mechanics, Aerospace Structures, High-Speed Aerodynamics, Propulsion, Thermodynamics

RELEVANT SKILLS

Programming/Software: SolidWorks CAD, CATIA, ANSYS (CFD simulations), MATLAB, Python, Delphi 10

Hardware/Technical: CNC Mill, CNC Lathe, 3D Printing, GD&T (ASME Y14.5-2018)

PROJECTS

Alabama Aerodynamics Experimental Research Operations (A.A.E.R.O.)

Co-Founder / CFD and Testing Lead | January 2025 – Present

- Lead design of drone propeller with vortex inducers to evaluate aerodynamic performance.
- Simulated propeller thrust and acoustic characteristics using Computational Fluid dynamics (CFD) in ANSYS and SolidWorks and compared with real-world test data.
- Performed Finite Element Analysis (FEA) on designs to assess structural integrity and material performance.
- Applied Geometric Dimensioning and Tolerancing (GD&T) for precise manufacturing of components.
- Conducted static test stand experiments measuring thrust, RPM, and acoustics for multiple design iterations.
- Utilized wind tunnel for aerodynamic data collection and model validation.

Alabama Rocketry Association – Liquids Team

Member | October 2023 – Present

- Collaborated on designing components for a single-stage liquid propellant rocket, focusing on fluids systems.
- Designed test stand and created P&ID diagrams adhering to safety standards.
- Performed hydrostatic proof testing of flight hardware to demonstrate tank compliance with MIL-STD-810G/SMC-S-016 qualification requirements.
- Utilized a CNC Lathe to manufacture a graphite nozzle within design tolerances.
- Assisted in setup and execution of multiple hot fire tests for LOX/Kerosene engine, including short-duration and duration tests at Tuscaloosa National Airport, analyzing performance data to identify areas for improvement.

Genesis Drone Simulation Project

Personal Project | August 2025 - Present

- Engineered a sophisticated quadcopter simulation in Python leveraging the Genesis physics engine, incorporating real-time aerodynamics, rigid-body dynamics, and control systems.
- Integrated PID controllers for altitude hold, attitude stabilization (roll, pitch, yaw), and position maintenance, tuned with gains derived from engineering principles to achieve critical damping ($\zeta \approx 0.7-1.0$) and sub-second settling times, preventing oscillations and flips.
- Implemented manual flight controls via Pygame for intuitive WASD/spacebar inputs, mapping to differential RPM adjustments on propellers, alongside automated modes for hover and leveling using quaternion-to-Euler conversions and vector rotations.
- Developed multi-environment support for parallel drone instances, benchmarked for performance (e.g., >1000 steps/sec on CUDA), with custom camera rendering (third-person and minimap views) and HUD overlays displaying RPM, velocity, thrust, drag, and Euler angles.
- Optimized for realism by incorporating gravitational forces, empirical thrust/drag calculations, and anti-windup in PIDs, demonstrating proficiency in control theory, numerical methods, and software.

WORK EXPERIENCE

Trinity Farm, Lakeland, TN

Ranch Hand | May 2022 – January 2024

- Provided daily care for horses, including feeding, grooming, and medical assistance, ensuring health and safety.
- Supported customers and trainers with inquiries and wound treatment, demonstrating communication skills.
- Maintained farm infrastructure through mowing, fence repairs, and troubleshooting machinery, honing mechanical problem-solving abilities transferable to engineering maintenance tasks.

National Tire and Battery (NTB), Magnolia, TX

Tire Technician | August 2021 – February 2022

- Organized warehouse stock and maintained clean vehicle bays to optimize workflow and improve safety.
- Advised customers on vehicle maintenance and repairs, building interpersonal and technical advisory skills.
- Executed routine maintenance tasks including oil changes, tire repairs, battery installations, and brake replacements.
- Assisted mechanics in advanced repairs, gaining hands-on experience in mechanical systems and diagnostics.